

Method of u -substitution (antiderivatives for chain rule problems)

Steps for the u -substitution process:

1. Group and factor (when appropriate) to create the substitution for u .
2. Define $u(x)$, $\frac{du}{dx}$, and the differential du .
3. Rewrite the part of integral that shows the du grouped term; adjust constants as needed.
4. Complete the substitution process and integrate to find the antiderivative in terms of u .
5. Back substitute to write the final antiderivative form and then check the result.

Hints for using u -substitution

1. Let u = complex arguments under roots or raised to powers.
2. Let u = expressions in the denominator.
3. Let u = expressions that are the arguments of trigonometric, exponential, or log functions.

1 - 5. Find the antiderivative of each of the following. Give only FINAL, completely simplified answers in the boxes. Show work below the answer boxes.

$$1. \int 6x(3x^2 - 5)^4 dx =$$

(ii) Define the substitution u and find du :

$u =$

$\frac{du}{dx} =$

$du =$

(iii) Rewrite the integral with the du term grouped; adjust constants as needed:

substitute in the du FIRST then check to see if the constants need to be adjusted.

$$\int 6x(3x^2 - 5)^4 dx =$$

Is the expression for du an exact match to a term in the integral?

Yes (substitution is direct) / No (manipulation of constants is required)

(iv) Perform the substitution and integrate:

(v) Back-sub and check:

$$2. \int \frac{2x}{\sqrt{3x^2-7}} dx$$

(i) Rewrite and group terms so that the integral so that the calculus operations can be performed:

$$\int \frac{2x}{\sqrt{3x^2-7}} dx$$

(ii) Define the substitution u and find du :

$u =$

$$\frac{du}{dx} =$$

$du =$

(iii) Rewrite the integral with the du term grouped; adjust constants as needed:

substitute in the du FIRST then check to see if the constants need to be adjusted.

$$\int \frac{2x}{\sqrt{3x^2-7}} dx$$

Is the expression for du an exact match to a term in the integral?

Yes (substitution is direct) / No (manipulation of constants is required)

(iv) Perform the substitution and integrate:

(v) Back-sub and check:

3. $\int \frac{10x^2}{(x^3-7)^4} dx$

check:

(i) Rewrite/group/factor:

(ii) Define u and du :(iii) Rewrite with du and adjusted constants: $\int \frac{10x^2}{(x^3-7)^4} dx =$ Is the expression for du an exact match to a term in the integral?

Yes (substitution is direct) / No (manipulation of constants is required)

(iv) - (v) Perform the substitution, integrate, back substitute and check.

4. $\int \frac{e^{2x}+1}{(e^{2x}+2x)^2} dx$

check:

(i) Rewrite/group/factor:

(ii) Define u and du :(iii) Rewrite with du and adjusted constants: $\int \frac{e^{2x}+1}{(e^{2x}+2x)^2} dx$ Is the expression for du an exact match to a term in the integral?

Yes (substitution is direct) / No (manipulation of constants is required)

(iv) - (v) Perform the substitution, integrate, back substitute and check.

5. $\int (12a + 3) \sin(4a^2 + 2a) \, da =$

(i) Rewrite/group/factor:

(ii) Define u and du :

(iii) Rewrite with du and adjusted constants:

Is the expression for du an exact match to a term in the integral?

Yes (substitution is direct) / No (manipulation of constants is required)

(iv) - (v) Perform the substitution, integrate, back substitute and check.

1 - 9. Find the antiderivative of each of the following. Do all work on separate loose leaf paper. Box only FINAL, completely simplified answers. All work must be show as in the previous examples and it must be neat, organized, and in correct mathematical language.

1. $\int \frac{5y^2}{(y^3+2)^3} dy$

2. $\int \frac{3x}{\sqrt{25-x^2}} dx$

3. $\int \cos(10x+1) dx$

4. $\int_1^2 6\theta^2 \sec \theta^3 \tan \theta^3 d\theta$

5. $\int \frac{4y^2}{(y^3+2)} dy$

6. $\int 7x^3 e^{x^4} dx$

7. $\int (6y^2 - 2)(y^3 - y)^4 dy$

8. $\int \frac{\sin x}{\cos^4 x} dx$

Remember, $\cos^4 x = (\cos x)^4$

9. $\int \frac{(\ln x)^2}{x} dx$

11. $\int \frac{5a-10}{a^2-4a} da$

12. $\int \csc^2\left(\frac{x}{4}\right) dx$

1. Use the information given to sketch the graph of the function $y = f(x)$ on the axes below. Show the correctly labeled sign charts. Draw a large graph that completely fills the axes space. Choose and clearly show an appropriate x-axis scale. Show all relevant points on the graph.

The function $f(x)$ is continuous on domain $D = (-\infty, \infty)$.

$$f(-2) = 8$$

$f(x)$ has no vertical or horizontal asymptotes.

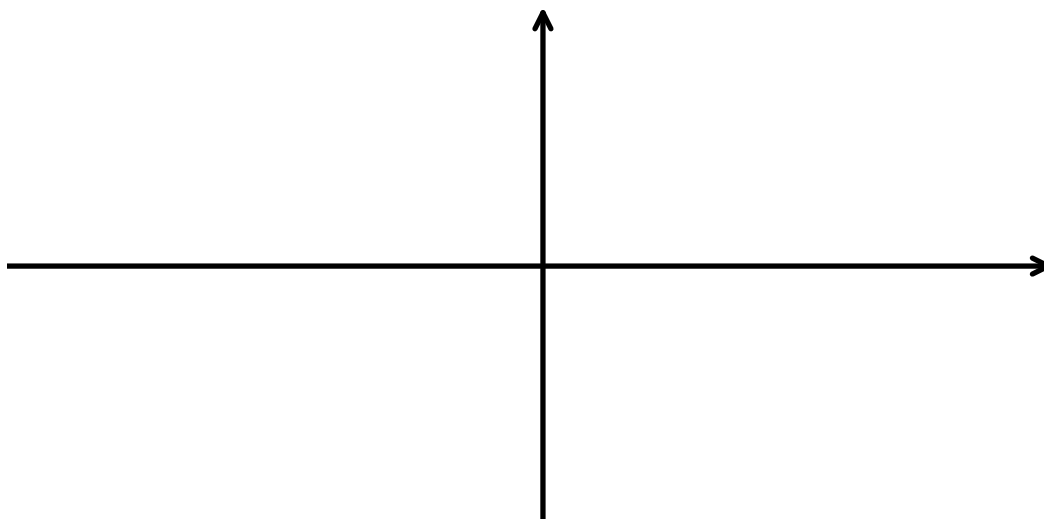
$f(x)$ is not differentiable at $x = -2$

$$f'(x) > 0 \text{ on } (-\infty, -2) \cup (2, 6)$$

$$f'(x) < 0 \text{ on } (-2, 2) \cup (6, \infty)$$

$$f''(x) > 0 \text{ on } (-2, 4)$$

$$f''(x) < 0 \text{ on } (-\infty, -2) \cup (4, \infty)$$



Recap: Using first and second order derivative information to draw graphs:

Use $y = f(x)$ to find all extreme and inflection points and additional points to graph.	$y = f(x_C)$ or $(x, y) = (x_C, f(x_C))$ $y = f(x_I)$ or $(x, y) = (x_I, f(x_I))$
Use $f'(x) = 0$ and $f'(x)$ d.n.e to find all the critical numbers. x_C .	If $f'(x) > 0$, then the function $f(x)$ is increasing. If $f'(x) < 0$, then the function $f(x)$ is decreasing.
Use $f''(x) = 0$ and $f''(x)$ d.n.e to find the x-values for the inflection points x_I .	If $f''(x) > 0$, then the graph of the function $f(x)$ is concave up. If $f''(x) < 0$, then the graph of the function $f(x)$ is concave down.

2. A function $f(x)$ has critical numbers at $x_C = -3$, $x_C = -1$, $x_C = 0$, and $x_C = 1$. The second order derivative is $f''(x) = 4x^3 + 9x^2 - 2x - 3$. Show how the Second order derivative test is used to determine whether each of the critical numbers locate a local maximum or minimum.

2. Use the second order derivative test to find and identify all the local extreme values for the function $f(x) = x^4 + 4x^3 + 4x^2 - 3$. Do not draw sign charts.

Absolute Extrema

Let $f(x)$ be defined on the interval (a,b) and let c be in the interval.

$f(c)$ is the absolute minimum of $f(x)$ on the interval if $f(x) \geq f(c)$ for all x in the interval.

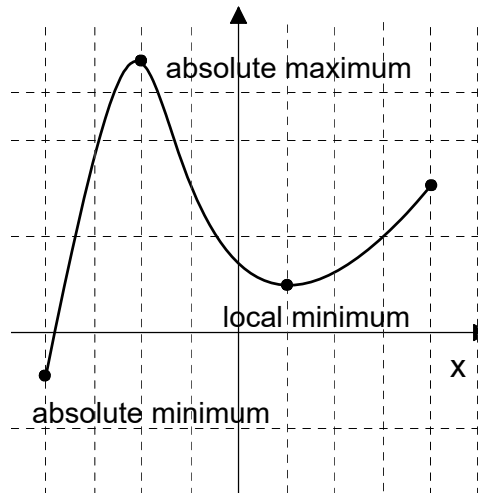
$f(c)$ is the absolute maximum of $f(x)$ on the interval if $f(x) \leq f(c)$ for all x in the interval.

The function $f(x)$ has an absolute extrema at c if it has either an absolute maximum or an absolute minimum at $x = c$.

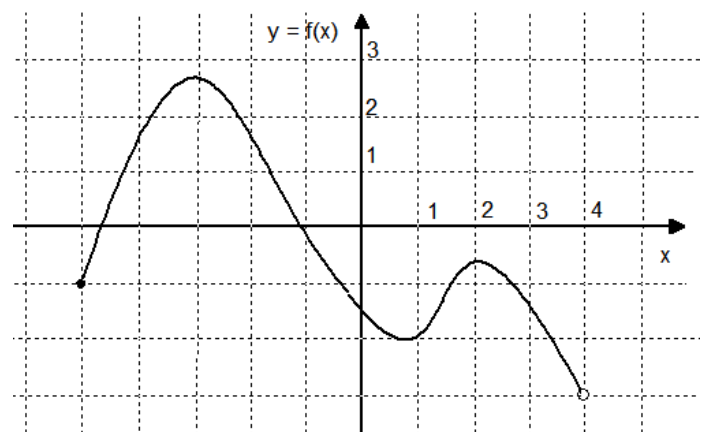
Extreme Value Theorem

A function $f(x)$ that is continuous on a closed interval $[a,b]$ will have both a absolute maximum and an absolute minimum on the interval.

Example:



1. The following graph represents a function defined on the indicated interval. Identify all the local and absolute extrema on the interval.



Steps for finding the *absolute* extrema for $f(x)$.

1. Use $f'(x)$ to find all the critical numbers for $f(x)$ in $[a,b]$: $f'(x) = 0$ **or** $f'(x)$ d.n.e.
2. Evaluate $f(x)$ for all critical numbers AND evaluate $f(x)$ at the endpoints a and b .
3. The absolute maximum value of $f(x)$ is the largest value found in step 2
The absolute minimum value of $f(x)$ is the smallest value found in step 2.

2 - 6. Find the absolute maximum value and the absolute minimum value, if any, for each of the following functions.

2. $h(x) = 2x^3 + 3x^2 - 12x + 4$ on $[-4, 2]$

step 1:

$h'(x) =$

critical numbers:

step 2: Evaluate $h(x_c)$ for all critical numbers x_c .

Evaluate $h(x)$ at endpoints.

x	$h(x)$

step 3: Identify absolute extrema.

3. $f(x) = x - 4 \ln x$ on $[1,5]$

$f'(x) =$

critical numbers:

Evaluate $f(x)$ for all critical numbers.

Evaluate $f(x)$ at endpoints.

x	$f(x)$

Identify all absolute extrema.

2 - 6. Find the absolute maximum value and the absolute minimum value, if any, for each of the following functions.

4. $g(t) = \frac{2t^2 + 3}{t^2 - 4}$ on $[-1, 2]$

critical numbers:

Evaluate $g(t)$ for all critical numbers.

Evaluate $g(t)$ at endpoints.

t	$g(t)$

Identify all absolute extrema.

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5. $g(x) = (x + 1)^4(2x - 5)^2$ on $[-2, 2]$

critical numbers:

$g'(x) =$

2 - 6. Find the absolute maximum value and the absolute minimum value, if any, for each of the following functions

5. $g(x) = (x + 1)^4 (2x - 5)^2$ on $[-2, 2]$
Evaluate $g(x)$ for all critical numbers.

x	$g(x)$

Evaluate $g(x)$ at endpoints.

Identify all absolute extrema.

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6. $f(x) = -3x^4 + 8x^3 + 18x^2 + 2$ on $[-2, 2]$

$f'(x) =$

critical numbers:

Evaluate $f(x)$ for all critical numbers.

x	$f(x)$

Evaluate $f(x)$ at endpoints.

Identify all absolute extrema.

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